PROTEINS-The Units of Life

Second letter

First letter	U	C	A	G	Third letter
U	UUU UUC Phenyl-alanine	UCU UCC UCA UCG Serine	UAU UAC Tyrosine	UGU UGC Cysteine	U
	UUA UUG Leucine		UAA Stop codon UAG Stop codon	UGA Stop codon UGG Tryptophan	C A G
C	CUU CUC CUA CUG Leucine	CCU CCC CCA CCG Proline	CAU CAC Histidine	CGU CGC CGA CGG Arginine	U
			CAA CAG Glutamine		C A G
A	AUU AUC AUA Isoleucine	ACU ACC ACA ACG Threonine	AAU AAC Asparagine	AGU AGC Serine	U
	AUG Methionine; start codon		AAA AAG Lysine	AGA AGG Arginine	C A G
G	GUU GUC GUA GUG Valine	GCU GCC GCA GCG Alanine	GAU GAC Aspartic acid	GGU GGC GGA GGG Glycine	U
			GAA GAG Glutamic acid		C A G

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Figure 12.5

Proteins play key roles in a living system

Three examples of protein functions

- Catalysis:
  Almost all chemical reactions in a living cell are catalyzed by protein enzymes.

- Transport:
  Some proteins transports various substances, such as oxygen, ions, and so on.

- Information transfer:
  For example, hormones.

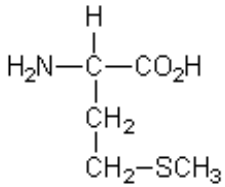
Alcohol dehydrogenase oxidizes alcohols to aldehydes or ketones

Haemoglobin carries oxygen

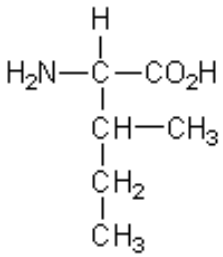
Insulin controls the amount of sugar in the blood

Amino acid structures

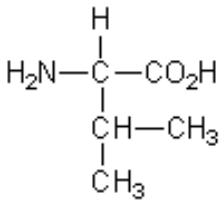
methionine (M)



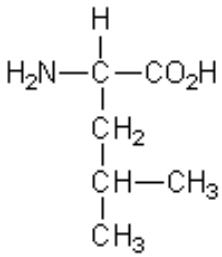
isoleucine (I)



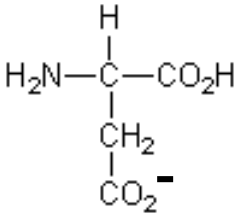
valine (V)



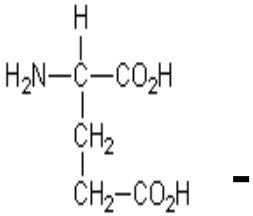
leucine (L)



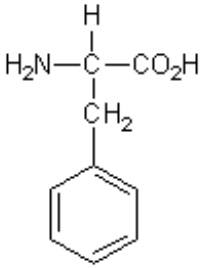
aspartic acid (D)



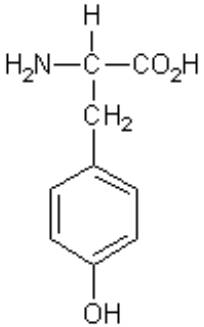
glutamic acid (E)



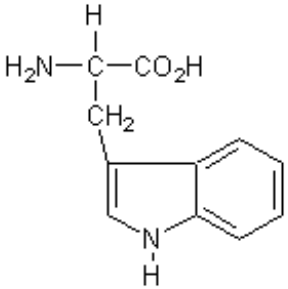
phenylalanine (F)



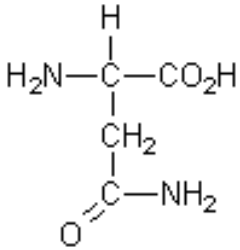
tyrosine (Y)



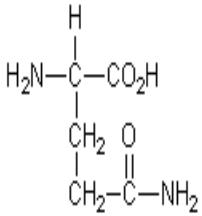
tryptophan (W)



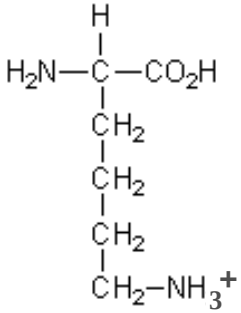
aspargine (N)



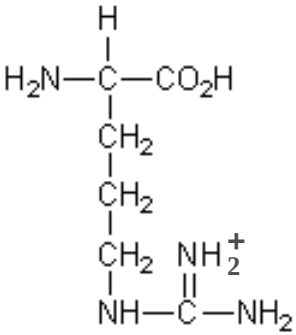
glutamine (Q)



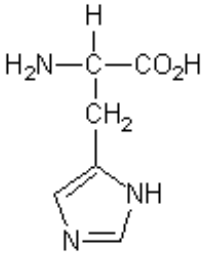
lysine (K)



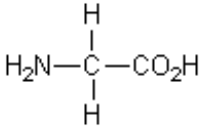
arginine (R)



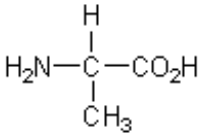
histidine (H)



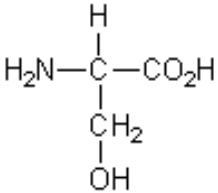
glycine (G)



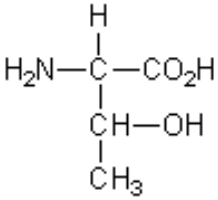
alanine (A)



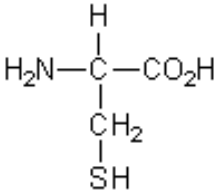
serine (S)



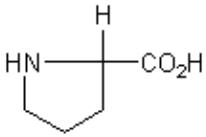
threonine (T)



cysteine

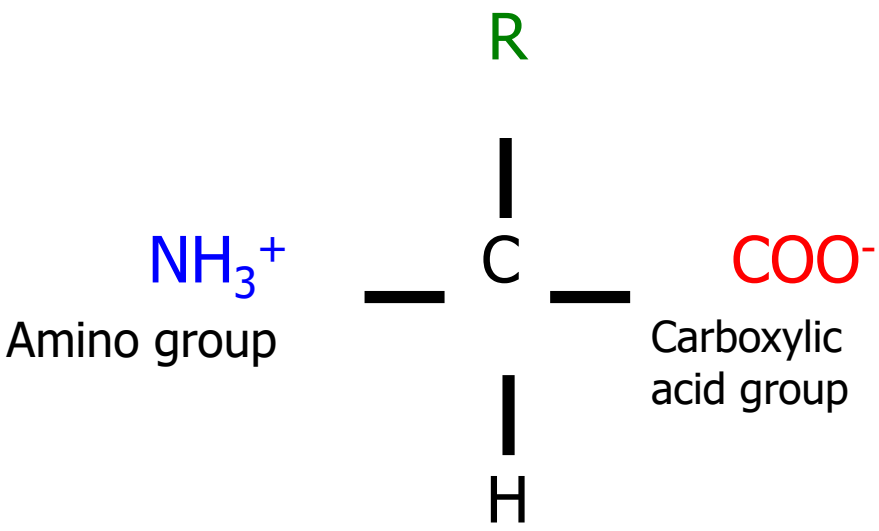


proline



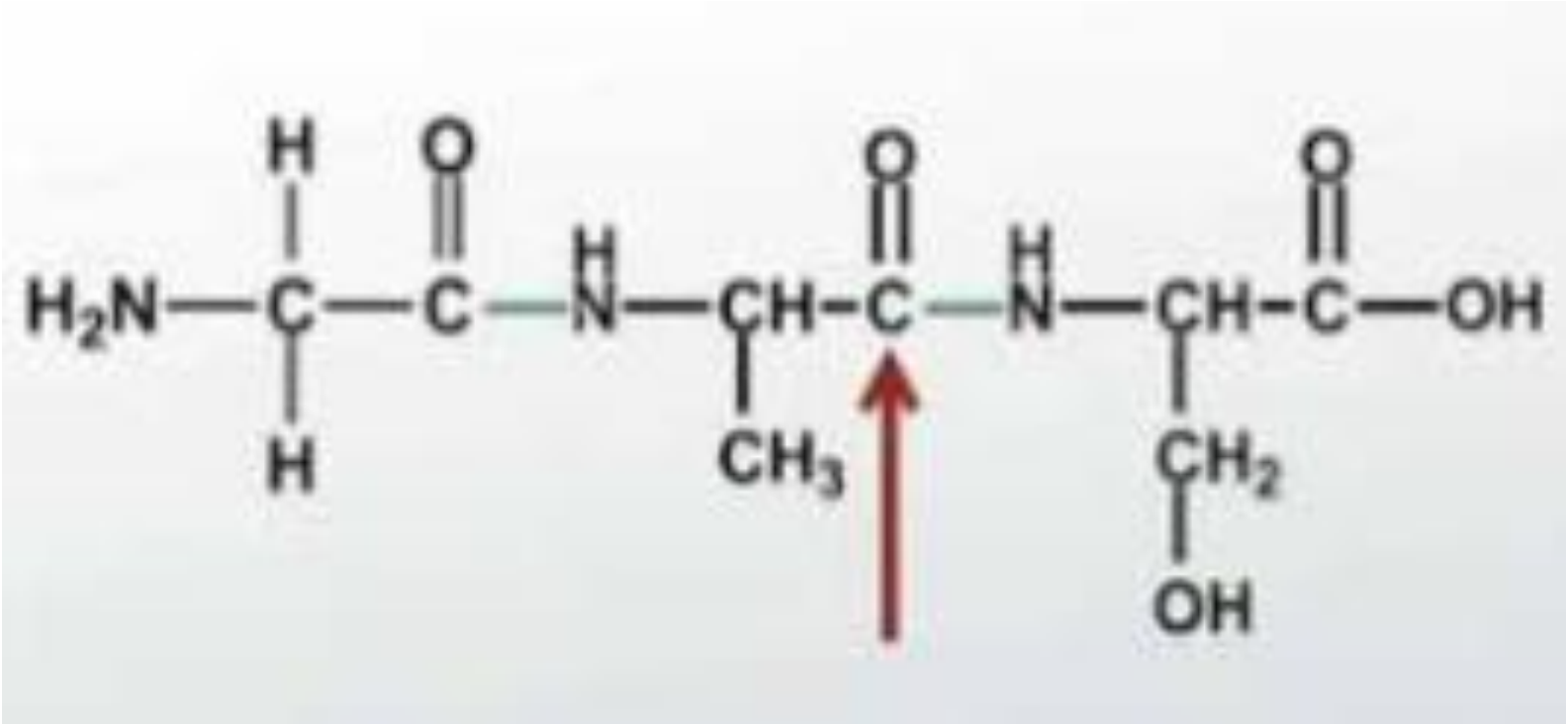
Amino acid: Basic unit of protein

Different side chains, R, determines the properties of 20 amino acids.



An amino acid

Peptide bond formation



©Alberts *et al*. (1998)

- You should know the structure of a polypeptide chain (protein)!

PRIMARY STRUCTURE

The sequence of amino acids

MIL1 sequence:

>gi|7662506|ref|NP_056182.1| MIL1 protein [Homo sapiens]

MEDCLAHLGEKVSQELKEPLHKALQMLLSQPVTYQAFRECTLETTVHASGWNKILVPLVLLRQML
LELTRLGQEPLSALLQFGVTYLEDYSAEYIIQQGGWGTVFSLESEEEEYPGITAEDSNDIYILPS
DNSGQVSPPESPTVTTSWQSESLPVSLSASQSWHTESLPVSLGPESWQQIAMDPEEVKSLDSNGA
GEKSENNSSNSDIVHVEKEEVPEGMEEAAVASVVLPARELQEALPEAPAPLLPHITATSLLGTRE
PDTEVITVEKSSPATSLFVELDEEEVKAATTEPTEVEEVVPALEPTETLLSEKEINAREESLVEE
LSPASEKKPVPPSEGKSRLSPAGEMKPMPLSEGKSILLFGGAAAVAILAVAIGVALALRKK

length: 386amino acids

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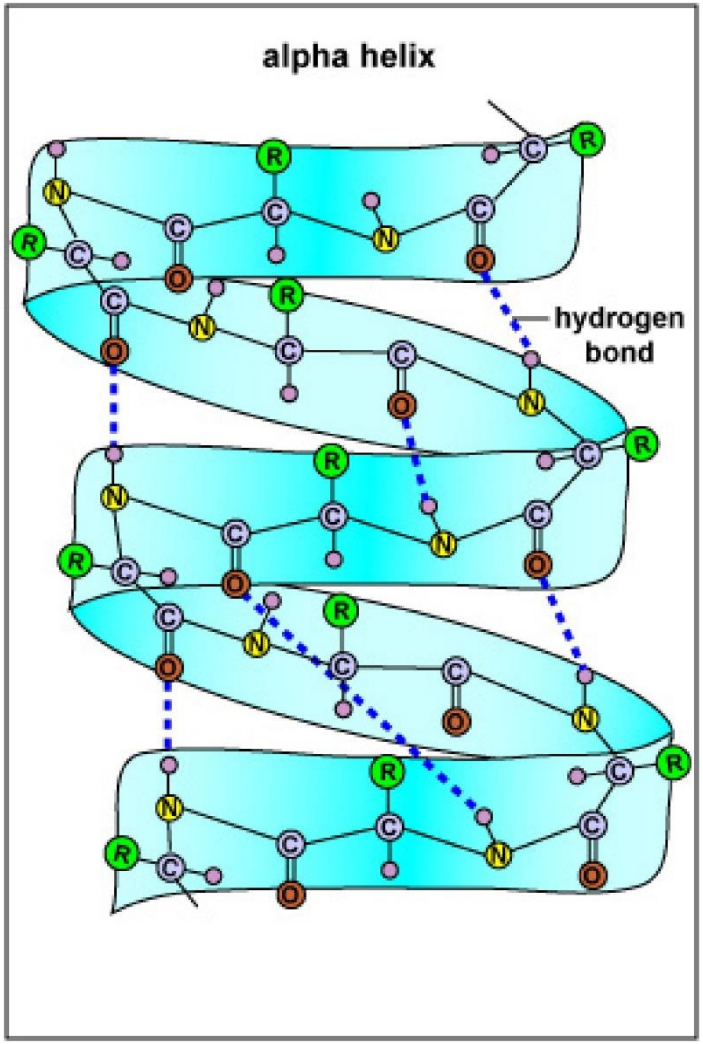
SECONDARY STRUCTURE

The folding of the N-C-C backbone of the polypeptide chain using weak hydrogen bonds



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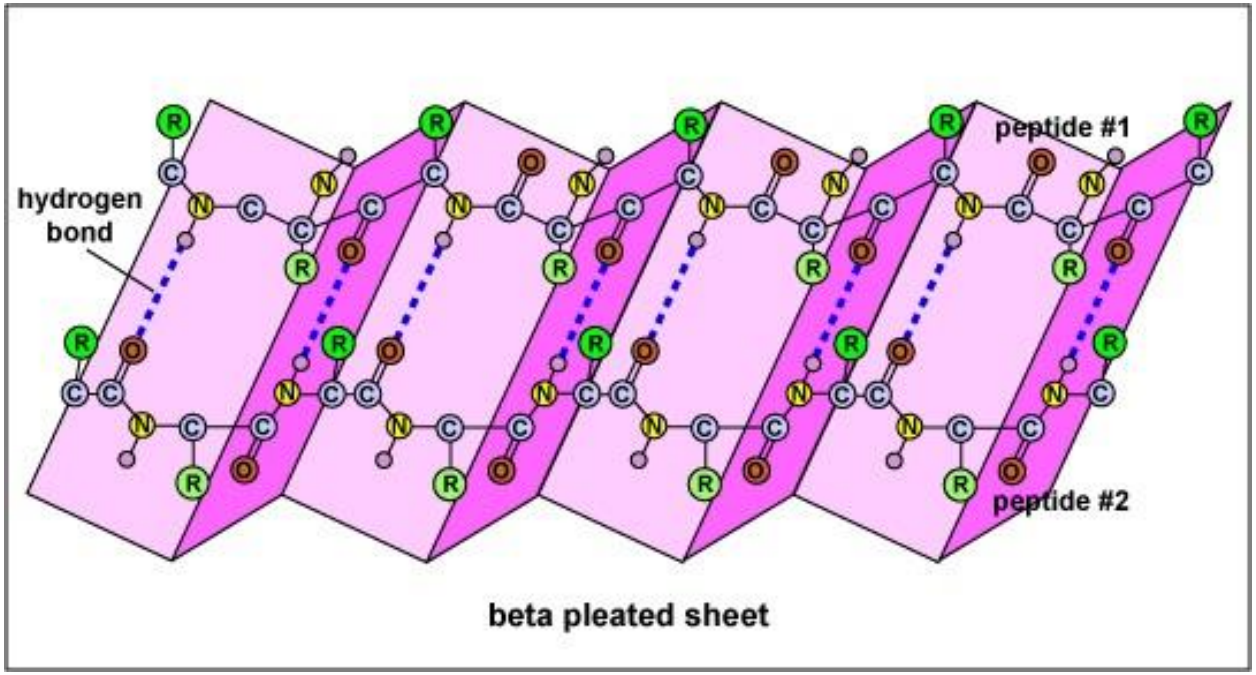
SECONDARY STRUCTURE

- This produces the **alpha helix** and **beta pleating**
- The length of the helix or pleat is determined by certain amino acids that will not participate in these structures
  (e.g. proline)



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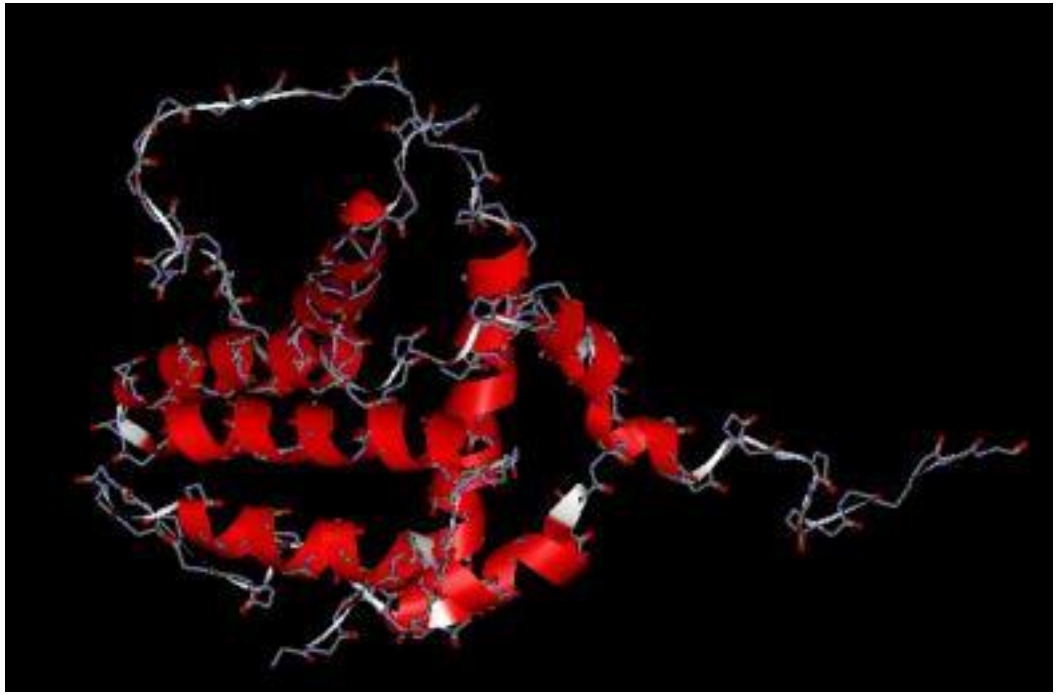
TERTIARY STRUCTURE

The folding of the polypeptide into domains whose chemical properties are determined by the amino acids in the chain

MIL1 protein



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TERTIARY STRUCTURE

- This folding is sometimes held together by strong covalent bonds
  (e.g. **cysteine-cysteine** disulphide bridge)
- Bending of the chain takes place at certain amino acids
  (e.g. proline)
- **Hydrophobic amino acids** tend to arrange themselves **inside** the molecule
- **Hydrophilic amino acids** arrange themselves on the **outside**

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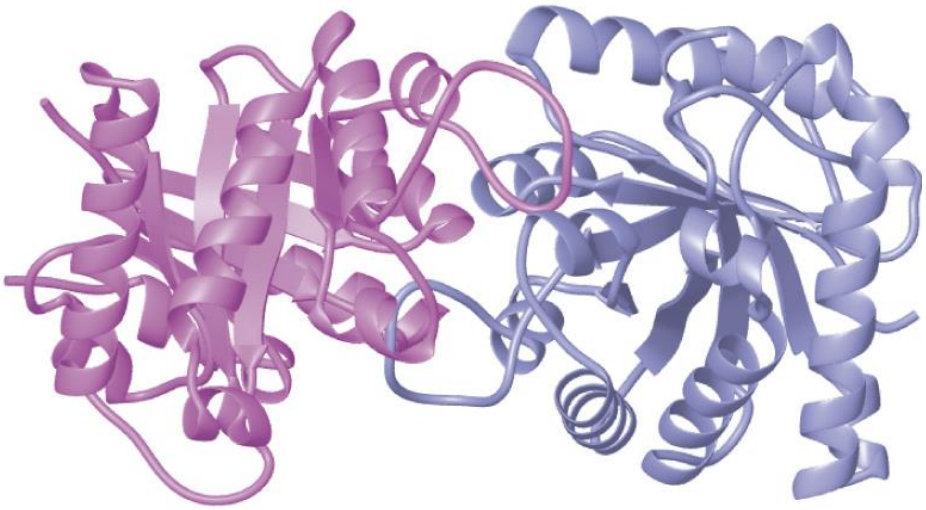
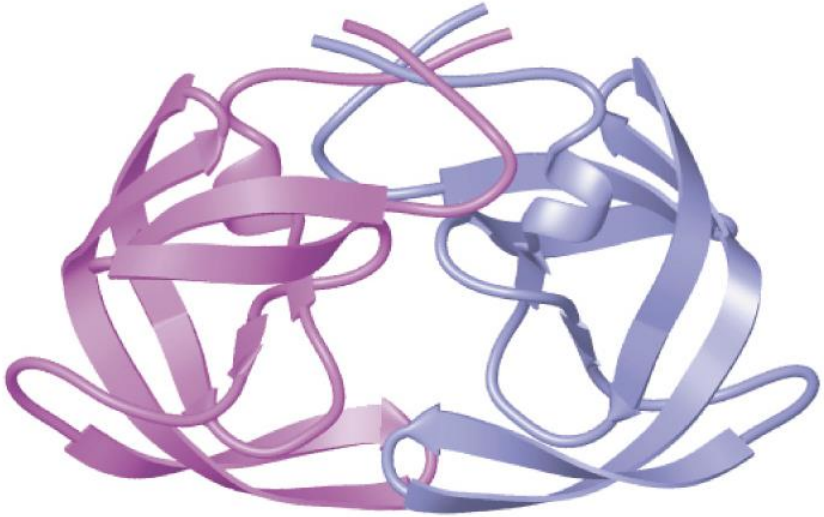
Quaternary structure of multidomain proteins

(a)



Chicken triose phosphate isomerase

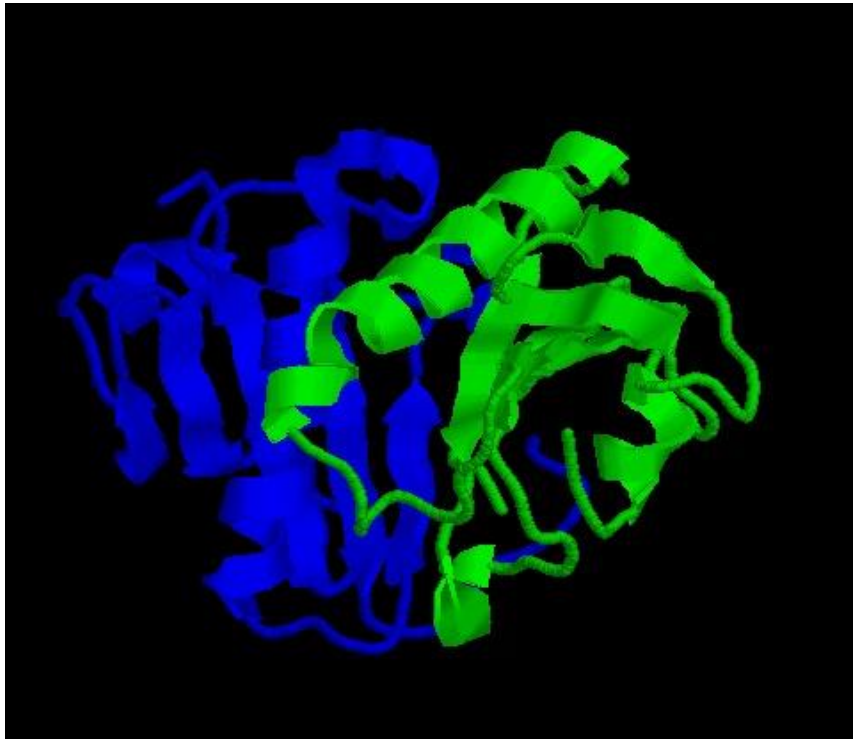
(b)



HIV-1 aspartic protease

QUATERNARY STRUCTURE

Some proteins are made of several polypeptide subunits (e.g. haemoglobin has four)



Protein Kinase C

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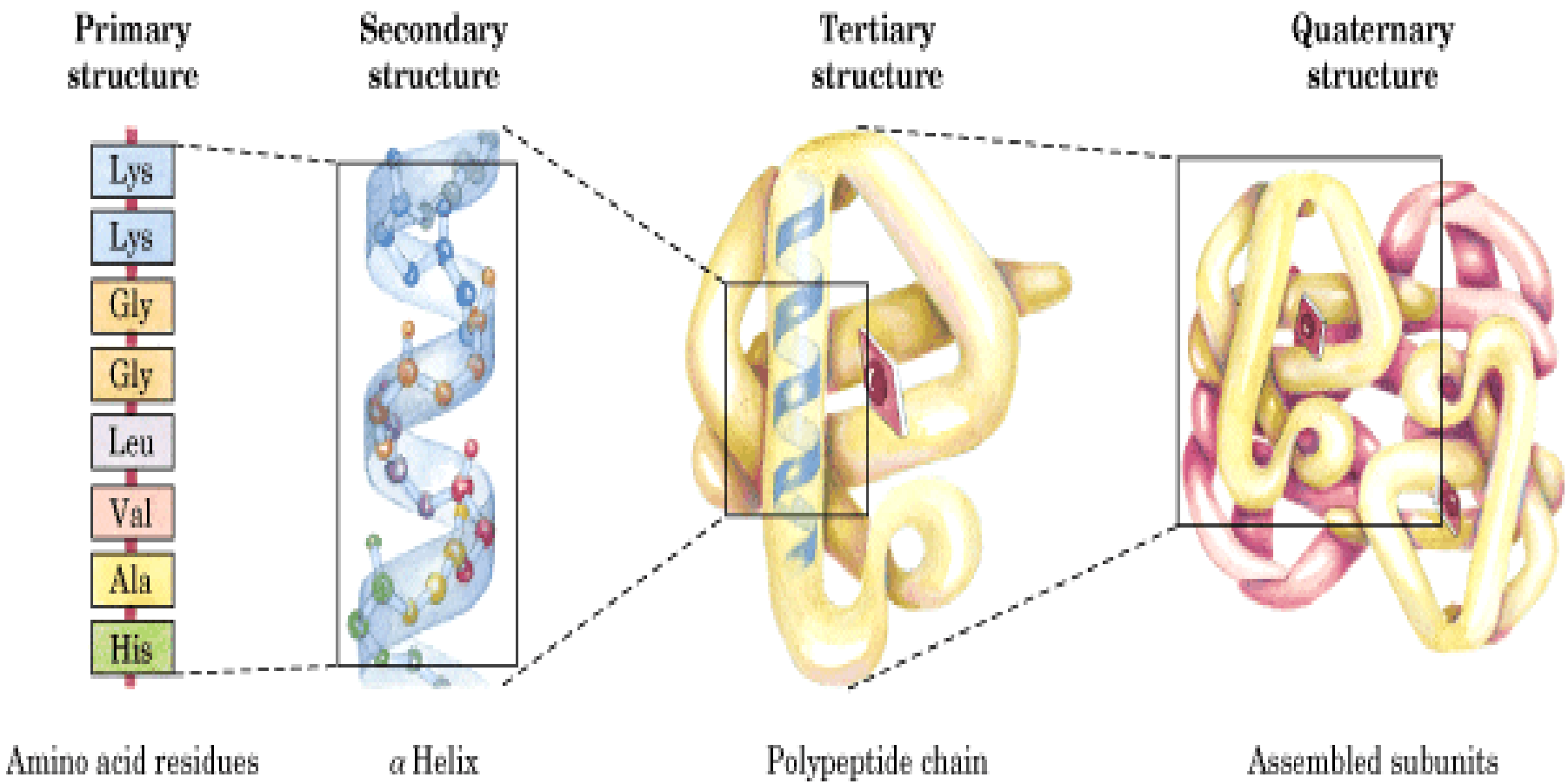
Protein structure: overview

Structural element	Description
primary structure	amino acid sequence of protein
secondary structure	helices, sheets, turns/loops
super-secondary structure	association of secondary structures
domain	self-contained structural unit
tertiary structure	folded structure of whole protein • includes disulfide bonds
quaternary structure	assembled complex (oligomer) • homo-oligomeric (1 protein type) • hetero-oligomeric (>1 type)


Primary structure
Secondary structure
Tertiary structure
Quaternary structure
Lys
Lys
Gly
Gly
Leu
Val
Ala
His
Amino acid residues
α Helix
Polypeptide chain
Assembled subunits